Chapter 3...

Student's t-Distribution

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Born in Canterbury, England to Agnes Sealy Vidal and Colonel Frederic Gosset, Gosset attended Winchester College before reading chemistry and mathematics at New College, Oxford. Upon graduating in 1899, he joined the brewery of Arthur Guinness & Son in Dublin, Ireland.

As an employee of Guinness, a progressive agro-chemical business, Gosset applied his statistical knowledge – both in the brewery and on the farm – to the selection of the best yielding varieties of barley. Gosset acquired that knowledge by study, by trial and error, and by spending two terms in 1906–1907 in the biometrical laboratory of Karl Pearson. Gosset and Pearson had a good relationship. Pearson helped Gosset with the mathematics of his papers, including the 1908 paper, but had little appreciation of their importance. The papers addressed the brewer's concern with small samples; biometricians like Pearson, on the other hand, typically had hundreds of observations and saw no urgency in developing small-sample methods.

Another researcher at Guinness had previously published a paper containing trade secrets of the Guinness brewery. To prevent further disclosure of confidential information, Guinness prohibited its employees from publishing any papers regardless of the contained information. However, after pleading with the brewery and explaining that his mathematical and philosophical conclusions were of no possible practical use to competing brewers, he was allowed to publish them, but under a pseudonym ("Student"), to avoid difficulties with the rest of the staff.[1] Thus his most noteworthy achievement is now called Student's, rather than Gosset's, t-distribution.

Contents ...

Key Words :

Students t-distribution, limiting distribution of t_n.

Objectives :

(1) To understand how t-distribution is derived from chi-square and standard normal variates.

(2) To study the properties of t-distribution.

(3) To note relationships between t distribution and Cauchy distribution.

3.0 Introduction

t-distribution is important due to its popularly used applications in statistical inference such as tests of hypothesis regarding mean, correlation coefficient, regression coefficients, determination of confidence interval for mean etc. Some of the applications will be discussed later in this book. William S. Gosset has published the paper on t-distribution in 1908 under the pseudonym 'student' hence it is called as Student's t-distribution.

3.1 P.D.F. of t-Distribution

Definition : Suppose U and V are independent random variables. If U follows N (0, 1) and V follows chi-square distribution with n degrees of freedom then $t = U/\sqrt{V/n}$ is said to follow t-distribution with parameter n, which is called as degrees of freedom. $\left[\text{i.e. } t = \dfrac{N(0,1)}{\sqrt{\chi^2_n/n}}\right]$.

We symbolically write it as t_n. of S. N variate with respect

Derivation of p.d.f. of t_n :

Since t is a transformation of U and V we find its p.d.f. as follows :

(i) Let $t = \dfrac{U}{\sqrt{V/n}}$, $Z = V$

Clearly, $-\infty < t < \infty$ and $0 < Z < \infty$

(ii) We get, $U = t\sqrt{\dfrac{Z}{n}}$ and $V = Z$

Hence, the Jacobian of transformation is

$$J = \frac{\partial (U, V)}{\partial (t, Z)} = \begin{vmatrix} \dfrac{\partial U}{\partial t} & \dfrac{\partial U}{\partial Z} \\[2mm] \dfrac{\partial V}{\partial t} & \dfrac{\partial V}{\partial Z} \end{vmatrix} = \begin{vmatrix} \sqrt{z/n} & \dfrac{t}{\sqrt{n}} \cdot \dfrac{1}{2\sqrt{z}} \\ 0 & 1 \end{vmatrix} = \sqrt{\frac{Z}{n}}$$

(iii) Joint p.d.f. of (t, Z) = Joint p.d.f. of (U, V) . |J|

$$g(t, Z) = f(u, v) \,.\, |J|$$

$$= f_1(u) \,.\, f_2(v)\, |J| \quad (\because \text{U and V are independent})$$

$$= \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \cdot \frac{1}{2^{n/2}\,\Gamma\frac{n}{2}} e^{-\frac{v}{2}} v^{n/2-1} \,.\, |J|$$

$$= \frac{1}{\sqrt{2\pi}\, 2^{n/2}\,\Gamma\frac{n}{2}} \,.\, e^{-\left(\frac{u^2}{2}+\frac{v}{2}\right)} \,.\, v^{\frac{n}{2}-1} \,.\, |J|$$

$$= \frac{1}{\sqrt{2\pi}\, 2^{n/2}\,\Gamma\frac{n}{2}} e^{-\frac{1}{2}\left(Z+\frac{t^2 z}{n}\right)} Z^{\frac{n}{2}-1} \cdot \sqrt{\frac{Z}{n}}$$

$$= \frac{1}{\sqrt{2\pi n}\, 2^{n/2}\,\Gamma\frac{n}{2}} e^{-\frac{z}{2}\left(1+\frac{t^2}{n}\right)} \,.\, Z^{\frac{n+1}{2}-1}$$

$$; z > 0, \; -\infty < t < \infty$$

(iv) To get p.d.f. of t we integrate g (t, Z) w.r.t. z,

$$p(t) = \int_0^\infty g(t, z)\, dz$$

$$p(t) = \frac{1}{\sqrt{2\pi n}\, 2^{n/2}\,\Gamma n/2} \cdot \int_0^\infty e^{-\frac{z}{2}\left(1+\frac{t^2}{n}\right)} \,.\, Z^{\frac{n+1}{2}-1}\, dz$$

$$p(t) = \frac{1}{\sqrt{2\pi n}\; 2^{n/2}\, \Gamma\frac{n}{2}} \cdot \frac{\Gamma\frac{n+1}{2}}{\left[\frac{1}{2}\left(1+\frac{t^2}{n}\right)\right]^{\frac{n+1}{2}}}$$

$$p(t) = \frac{1}{\sqrt{n}} \cdot \frac{\Gamma\frac{n+1}{2}}{\sqrt{\pi}\,\Gamma\frac{n}{2}} \cdot \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} \qquad \left(\because \sqrt{\pi} = \Gamma\frac{1}{2}\right)$$

$$p(t) = \frac{1}{\sqrt{n}} \cdot \frac{1}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}, \qquad -\infty < t < \infty$$

Thus, we can define a t-distribution with n degrees of freedom as a continuous r.v. having p.d.f.

$$f(t) = \frac{1}{\sqrt{n}} \cdot \frac{1}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}; \qquad -\infty < t < \infty$$

Note :

(1) It can be easily verified that $f(-t) = f(t)$, hence $f(t)$ is a symmetric function of t around 0.

(2) We symbolically write a r.v. follows t-distribution with n degrees of freedom as t_n.

3.2 Moments of t_n

(a) Mean $= E(T) = \int_{-\infty}^{\infty} t\, f(t)\, dt = \int_{-\infty}^{\infty} \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{t}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt$

Let, $g(t) = \frac{t}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}$.

Since, $g(-t) = -g(t)$, it is a odd function of t.

Hence $\int_{-\infty}^{\infty} g(t)\, dt = 0$

$\therefore$ Mean $= E(t) = \mu_1' = 0$.

(b) Central moments of order 2r, (μ_{2r}) [Even ordered central moments] ($r = 1, 2 \ldots$)

Since, $\mu_1' = 0$, raw moments and central moments are same.

$$\therefore \quad \mu_{2r} = E\left(t^{2r}\right) = \int_{-\infty}^{\infty} t^{2r} f(t)\, dt$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{t^{2r}}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt$$

Since, the integrand is even function we write

$$\mu_{2r} = \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot 2\int_{0}^{\infty} \frac{t^{2r}}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt$$

Making substitution $\frac{t^2}{n} = y$ we get,

$$t = \sqrt{ny}, \quad 0 < y < \infty \quad \text{and} \quad dt = \frac{\sqrt{n}}{2\sqrt{y}} dy$$

$$\therefore \quad \mu_{2r} = \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{2\sqrt{n}}{2} \int_{0}^{\infty} \frac{n^r y^r}{(1+y)^{\frac{n+1}{2}}} \frac{dy}{\sqrt{y}}$$

$$= \frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \int_{0}^{\infty} \frac{y^{r+\frac{1}{2}-1}}{(1+y)^{\frac{n+1}{2}}} dy$$

$$= \frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot B\left(r+\frac{1}{2}, \frac{n}{2}-r\right)$$

(for $r < n/2$); $r = 1, 2, 3, \ldots$

(i) If $r = 1$, then we get (for $n > 2$)

$$\text{Variance} = \mu_2 = n \frac{B\left(\frac{3}{2}, \frac{n}{2}-1\right)}{B\left(\frac{1}{2}, \frac{n}{2}\right)} = n \frac{\Gamma\frac{3}{2}\, \Gamma\left(\frac{n}{2}-1\right)}{\Gamma\frac{1}{2}\, \Gamma\frac{n}{2}}$$

$$= n\,\frac{1/2}{(n/2-1)} = \frac{n}{n-2} \quad (\text{if } n>2)$$

Comment : $\mu_2 > 1$

(ii) If $r = 2$, then for $n > 4$ we get,

$$\mu_4 = n^2 \frac{B\left(\frac{5}{2}, \frac{n}{2}-2\right)}{B\left(\frac{1}{2}, \frac{n}{2}\right)} = n^2 \frac{\Gamma\frac{5}{2}\,\Gamma\left(\frac{n}{2}-2\right)}{\Gamma\frac{1}{2}\,\Gamma\frac{n}{2}}$$

$$= n^2 \frac{\frac{3}{2}\cdot\frac{1}{2}}{\left(\frac{n}{2}-1\right)\left(\frac{n}{2}-2\right)} = \frac{3\,n^2}{(n-2)\,(n-4)} \quad (\text{if } n>4)$$

(c) Central moments of order 2r + 1 (μ_{2r+1}) ($r = 1, 2, \ldots$)

[Odd ordered central moments]

$$\mu_{2r+1} = E\left(T^{2r+1}\right) = \int_{-\infty}^{\infty} t^{2r+1}\, f(t)\, dt$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \frac{t^{2r+1}}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}\, dt$$

Since, the integrand is odd function the integral is zero.

$\therefore \quad \mu_{2r+1} = 0 \quad$ for $r = 1, 2, 3, \ldots$

Thus, all the moments of odd order are zero.

Therefore, first four central moments are.

$$\mu_1 = 0,\quad \mu_2 = \frac{n}{n-2},\quad \mu_3 = 0,\quad \mu_4 = \frac{3n^2}{(n-2)\,(n-4)}$$

$\therefore$ Coefficients of skewness $\beta_1 = 0$, $\gamma_1 = 0$, hence the distribution is symmetric about mean (zero).

Also, the coefficient of kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\,n^2}{(n-2)\,(n-4)}\,\frac{(n-2)^2}{n^2} = \frac{3\,(n-2)}{n-4}$$

$$= \frac{3\,(n-4+2)}{n-4} = 3 + \frac{6}{n-4} > 3$$

and $\quad \gamma_2 = \beta_2 - 3 = \frac{6}{n-4} > 0$, hence the distribution is leptokurtic.

As $n \to \infty$, $\beta_2 \to 3$, hence the distribution is asymptotically mesokurtic.

Nature of probability curve

The probability curve is symmetric about 0 and bell shaped. Hence we get, mean = mode = median = 0. For large n the curve will resemble to that of standard normal curve.

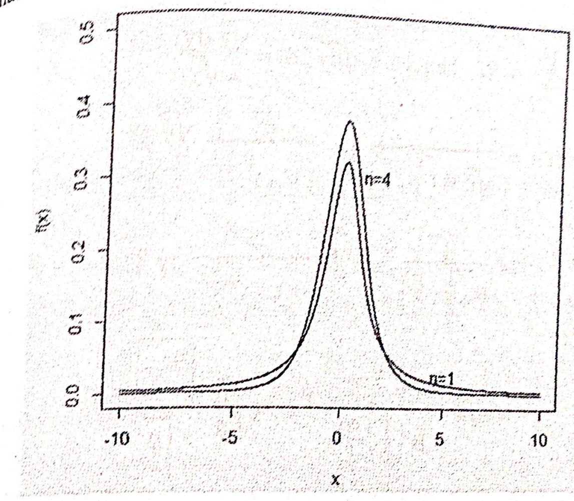

Fig. 3.1 : t-distribution probability curve using R software

Solved Examples

Example 3.1 : If a r.v. follows t-distribution with n degrees of freedom, then show that the mean deviation about mean is $\sqrt{\frac{n}{\pi}}\,\frac{\Gamma\frac{n-1}{2}}{\Gamma\frac{n}{2}}$

Solution Note that mean of t-distribution is zero, hence mean deviation about mean is

$$E(|T|) = \int_{-\infty}^{\infty} |t|\,.\,f(t)\, dt$$

Since the integrand is even function

$$E(|T|) = 2\int_0^{\infty} |t|\, f(t)\, dt$$

$$= 2\int_0^{\infty} \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{t}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}\, dt$$

put, $\frac{t^2}{n} = y$, hence $t = \sqrt{ny}$, $dt = \frac{\sqrt{n}\, dy}{2\sqrt{y}}$

$$\therefore \quad E(|T|) = \frac{2}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \int_0^{\infty} \frac{\sqrt{ny}}{(1+y)^{\frac{n+1}{2}}} \cdot \frac{\sqrt{n}\, dy}{2\sqrt{y}}$$

$$= \frac{1}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \sqrt{n} \int_0^{\infty} \frac{1}{(1+y)^{\frac{n+1}{2}}}\, dy \quad [1 = y^0 = y^{1-1}]$$

$$= \sqrt{n}\, \frac{B\left(1, \frac{n-1}{2}\right)}{B\left(\frac{1}{2}, \frac{n}{2}\right)} = \sqrt{\frac{n}{\pi}} \cdot \frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \qquad (\text{if } n > 1)$$

Example 3.2 : If μ_{2r} is central moment of order 2r of t-distribution with n d.f., then show that

$$\mu_{2r} = \frac{n(2r-1)}{n-2r}\, \mu_{2r-2} \; ; \; r < \frac{n}{2}$$

Solution : Note that,

$$\mu_{2r} = E(t^{2r}) = n^r \frac{B\left(r+\frac{1}{2}, \frac{n}{2}-r\right)}{B\left(\frac{1}{2}, \frac{n}{2}\right)}$$

$$\therefore \quad \mu_{2r-2} = n^{r-1} \cdot \frac{B\left(r-1+\frac{1}{2}, \frac{n}{2}-r+1\right)}{B\left(\frac{1}{2}, \frac{n}{2}\right)}$$

$$\therefore \quad \frac{\mu_{2r}}{\mu_{2r-2}} = n \cdot \frac{B\left(r+\frac{1}{2}, \frac{n}{2}-r\right)}{B\left(r-\frac{1}{2}, \frac{n}{2}-r+1\right)}$$

$$= n \cdot \frac{\Gamma\left(r+\frac{1}{2}\right)\Gamma\left(\frac{n}{2}-r\right)}{\Gamma\left(r-\frac{1}{2}\right)\Gamma\left(\frac{n}{2}-r+1\right)} = n \cdot \frac{\left(r-\frac{1}{2}\right)}{\left(\frac{n}{2}-r\right)} = n \cdot \frac{(2r-1)}{(n-2r)}$$

$$\therefore \quad \mu_{2r} = n \cdot \frac{(2r-1)}{n-2r} \cdot \mu_{2r-2}$$

Deduction : Note that $\mu_2 = \frac{n}{n-2}$, the relation for r = 2 gives

$$\mu_4 = n \cdot \left(\frac{4-1}{n-4}\right) \mu_2 = \frac{3n}{n-4} \cdot \frac{n}{n-2} = \frac{3n^2}{(n-2)(n-4)} \text{ if } n > 4.$$

3.3 Limiting distribution of t_n

Statement : If a r.v. follows t-distribution with n degrees of freedom, then as $n \to \infty$ the probability distribution of T tends to N(0, 1).

Proof : We quote the results required in obtaining the limiting probability distribution.

Result (1) $\frac{\Gamma(n+k)}{\Gamma(n)} \approx n^k$ for large n

Result (2) $\lim\limits_{n \to \infty} \left(1+\frac{c}{n}\right)^n = e^c$, for c constant

$$\lim_{n \to \infty} f(t) = \lim_{n \to \infty} \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \times \frac{1}{\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}}$$

$$= \lim_{n \to \infty} \frac{1}{\sqrt{n}} \cdot \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{n}{2}\right)} \times \frac{1}{\left(1+\frac{t^2}{n}\right)^{n/2}} \times \frac{1}{\left(1+\frac{t^2}{n}\right)^{1/2}}$$

$$= \lim_{n \to \infty} \frac{1}{\sqrt{n}} \times \frac{1}{\sqrt{\pi}} \lim_{n \to \infty} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \times$$

$$\lim_{n \to \infty} \frac{1}{\left[\left(1 + \frac{t^2}{n}\right)^n\right]^{1/2}} \times \frac{1}{\left(1 + \frac{t^2}{n}\right)^{1/2}}$$

$$= \lim_{n \to \infty} \frac{1}{\sqrt{\pi}} \frac{1}{\sqrt{n}} \cdot \left(\frac{n}{2}\right)^{\frac{1}{2}} \times \frac{1}{e^{t^2/2}}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \text{ which is the p.d.f. of } N(0, 1).$$

Hence the proof.

Note : (1) For $n = 1$, we get

$$f(t) = \frac{1}{\pi} \cdot \frac{1}{1 + t^2} \; ; \qquad -\infty < t < \infty$$

which is called as Cauchy distribution.

(2) As $n \to \infty$, t-distribution tends to $N(0, 1)$. However, the approximation holds good for $n > 30$.

Solved Examples

Example 3.3 : If $X_1, X_2 \ldots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then show that $Y = \frac{\bar{X} - \mu}{s / \sqrt{n}}$ follows t-distribution with $n - 1$ degrees of freedom where $s^2 = \frac{1}{n-1} \Sigma (X_i - \bar{X})^2$.

Solution : Note that $\bar{X}$ and $\frac{\Sigma (X_i - \bar{X})^2}{\sigma^2}$ are independent random variables. Moreover $\bar{X} \to N(\mu, \sigma^2/n)$ and $\Sigma (X_i - \bar{X})^2/\sigma^2 \to \chi^2_{n-1}$.

Let, $\quad U = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

Thus $U \to N(0, 1)$ and $V \to \chi^2_{n-1}$. Hence $Y = \frac{U}{\sqrt{V/(n-1)}} \to$ t-distribution with $n - 1$ d.f.

$$\therefore \quad Y = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \times \frac{1}{\sqrt{\frac{\Sigma (X_i - \bar{X})^2}{(n-1)\sigma^2}}} = \frac{(\bar{X} - \mu)\sqrt{n}}{\sigma} \times \frac{\sigma}{s}$$

$$Y = \frac{\bar{X} - \mu}{s/\sqrt{n}} \text{ follows t-distribution with } (n-1) \text{ d.f.}$$

Example 3.4 If t follows Student's t-distribution with 15 d.f. find

(i) c such that $P(-c \le t \le c) = 0.8$

(ii) k such that $P(t^2 \ge k) = 0.05$

(iii) a such that $P(t \le a) = 0.1$

Solution : From statistical table we get $P(|t| \ge c)$

(i) $\quad P(-c \le t \le c) = P(|t| \le c) = 0.8$

$\therefore \quad P(|t| > c) = 0.2,$

From statistical table we get, $P(|t_{15}| > 1.341) = 0.2$

$\therefore \quad C = 1.341$

(ii) $\quad P(t^2 \ge k) = P(|t_{15}| \ge \sqrt{k}) = 0.05$

From statistical table we get,

$P(|t_{15}| > 2.131) = 0.05$, therefore $\sqrt{k} = 2.131$

$\therefore \quad k = 4.5412$

(iii) $\quad P(t \le a) = 0.1 \quad a < 0$

$P(t \ge -a) = 0.1$. Due to symmetry.

$\therefore \quad P[|t| \ge -a] = 0.2$

From statistical tables,

$P(|T_{15}| > 1.341) = 0.2$

$\therefore \quad a = -1.341$

COMPUTATION OF PROBABILITIES FOR t-DISTRIBUTION

Compute $P[t_{15} > 1.341]$ using :

(A) Statistical tables (B) MS-Excel (C) R-software.

(A) Statistical table gives $P(|t_{15}| > |1.34|) = 0.2$

$P(t_{15} > 1.341) + P(t_{15} < -1.341) = 0.2$

$P(t_{15} > 1.341) = P(t_{15} < -1.341) = 0.1.$

Due to symmetry $P(t_{15} > 1.34) = P(t_{15} < -1.341) = 0.1.$

(B) Using MS-EXCEL we find $P(t_{15} > 1.341)$ as follows :

Step 1 : Click on $\boxed{\text{Insert}}$ on the Excel sheet.

Points to Remember

1. Suppose U follows N (0, 1) and V follows chi-square distribution with n degrees of freedom. If U and V are independent random variables then $t = \dfrac{U}{\sqrt{v/n}}$ follows t-distribution with n degrees of freedom.

2. The p.d.f. of t-distribution with n degrees of freedom is

$$f(t) = \frac{1}{\sqrt{n}} \cdot \frac{1}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}} \; ; \; -\infty < t < \infty$$

3. If $T \rightarrow t_n$ with n degrees of freedom then

   (i) $E(T) = \text{mean} = 0$, $\text{Var}(T) = \mu_2 = \dfrac{n}{n-2}$.

(ii) All odd ordered central moments are zero i.e. $\mu_{2r+1} = 0$ for r = 2, 3 .... .

(iii) All even ordered central moments are given by

$$\mu_{2r} = \frac{n^r}{B\left(\frac{1}{2}, \frac{n}{2}\right)} B\left(r + \frac{1}{2}, \frac{n}{2} - r\right) \left(\text{for } \frac{n}{2} > r\right) \quad r = 1, 2 \ldots$$

(iv) $\mu_4 = \dfrac{3n^2}{(n-2)(n-4)}$, $\beta_1 = 0$, $\gamma_1 = 0$, $\beta_2 = 3 + \dfrac{6}{n-4}$, $\gamma_2 = \dfrac{6}{n-4}$

(v) The mean deviation about mean $= \sqrt{\dfrac{n}{\pi}} \dfrac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}$.

(vi) As $n \to \infty$ the probability distribution of T tends to N (0, 1).

4. If μ_{2r} is central moment of order 2r of t-distribution with n d.f. then

$$\mu_{2r} = \frac{n(2r-1)}{n-2r} \mu_{2r-2}; \quad r < \frac{n}{2}$$

5. If $X_1, X_2, \ldots, X_n$ is a random sample from N (μ, σ^2) then $Y = \dfrac{\bar{X} - \mu}{s/\sqrt{n}}$ follows t-distribution with (n – 1) degrees of freedom where $s^2 = \dfrac{1}{n-1} \Sigma (X_i - \bar{X})^2$.

6. t-distribution is symmetric about 0.

7. If a r.v. follows t distribution with 1 degree of freedom we get $f(t) = \dfrac{1}{\pi} \cdot \dfrac{1}{1+t^2}$; $-\infty < t < \infty$ which is called Cauchy distribution.

8. If $X_1, X_2, \ldots, X_{n_1}$ is a random sample from N (μ_1, σ^2) and $Y_1, Y_2, \ldots, Y_{n_2}$ is a random sample from N (μ_2, σ^2) [i.e. $\sigma_1^2 = \sigma_2^2 = \sigma^2$] then

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\chi^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \to \text{students t distribution with } n_1 + n_2 - 2 \text{ d.f.}$$

where $\chi^2 = \dfrac{[\Sigma (X_i - \bar{X})^2 + \Sigma (Y_i - \bar{Y})^2]}{n_1 + n_2 - 2}$